Enhancing the Experience Of Music-Ritual through Gesturally-Controlled Interactive Technology

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Abstract

This paper presents the possibility of applying modern gesturally-controlled interactive music technology to the aesthetic framework of "music-ritual", with the aim of enhancing the experience for the audience and participants by more directly mapping gesture with sound production. The potential of this kind of application could be to increase the communicability of symbolic meaning through concept, sound, and gesture.

1 Introduction

The concept of "music-ritual" has been explored and utilised extensively by many Western "art music" composers during the 20th Century, particularly during the 1960s and 1970s, and was to a large extent motivated by the social and artistic "disillusionment" of intellectual left-wing at the time. My own interest in these ideas has been particularly focussed on how different physically expressive forms (such as movement in space and gesture) can be used in conjunction with sound to enhance the experience for both the performer and the audience. This stemmed from my activities as an instrumental composer, which for some time have been focussed on how meaning can be communicated by sound. These works made use of various methods, applying concepts derived from psychoacoustics, symbolism and ritual, and were often inspired by composers such as Karlheinz Stockhausen and Pauline Oliveros.

In 1999 I began composing electroacoustic music (primarily using Pro Tools to treat and mix found sounds), which I believe has given me much more creative freedom than in purely notated music, as it has allowed me to manipulate the spatial and textural qualities of the sound itself. This kind of work can induce a feeling of complete immersion and a sense of physical "disembodiment", thus allowing both the composer and listener to explore a much deeper, internalised level of musical communication.

My most recent works have begun to make use of both ritualistic forms and electronically modified sound, as it is my contention that by using currently available gestural and movement technology as the medium of control, these two forms of creative expression could be combined, and the realization of ritual music forms could be further enhanced.

The ability of this kind of technology to accurately map movement to sound or sound manipulation makes it ideal for directly representing sound in movement. This means that a meaningful connection needs to be made between the two forms of expression, so that they can be experienced as two parts of the one whole.

This research is still in its formative stages, and has yet to draw any definite conclusions. This paper presents an outline of my background research, my ideas, and provides a model for the actual realization of these ideas. There are 3 stages to this discussion:

Background Research. I will begin by briefly discussing the concept of sound as a communicator of extra-musical ideas, with particular reference to Tibetan Buddhist traditions. These concepts will also be examined within the context of 20th Century art music, and will examine some of the relevant work of Karlheinz Stockhausen and Pauline Oliveros.

New Media and Methods. I will then examine some of the new aesthetic concepts brought forward by new gestural and movement controller technology, with particular reference to the work of Atau Tanaka and Tod Machover.

Conclusion. This section consolidates the various points presented throughout the discussion.

2 Background Research

2.1 Theory, Physics and Metaphysics

Essentially, my interest in the connection between gesture and sound is based on the belief that music – the organised relationships and interactions in time between different pitches and rhythms – reflects the organised relationships and interactions between the different elements of other systems within the universe, and hence the universe itself. Through the numerical symbolism of music, the interplay between forces that would normally lie beyond the realm of the human senses becomes something experiential. "In musical experience we come into direct contact with these principles [of numerical symbolism]; the connection between physical reality and metaphysical principles can be felt in music as nowhere else" (Daniélou 1943, p. 1).

Some ancient texts (e.g. the Indian *Vedas*) suggest that sound is in fact a *direct* – not merely symbolic – representation of these universal interplays, implying that each object, be it an atom or a star, has a vibrational frequency and thus a sound. Although these sounds are normally too subtle to be heard, audible sounds can be made that correspond directly to these more subtle vibrations. This belief is strongly held by Stockhausen, who has referred to it many times in his writings and conversations. "Every object in the world, down to the smallest atom, produces waves which can be transformed into acoustic waves. […] Everything makes sounds" (Stockhausen 1971, cited in Cott 1974, p. 76). In China, music was used as a tool for the diagnosis and rectification of social or political disturbances (Campbell 1959, p. 453).

The belief that sound is both a physically and psychologically active force is one that has found wide application in electronic and computer-generated music. For many years, electronic music artists such as Steven Halpern have explored these theories, and believe that by correctly applying audible sounds that correspond to more subtle vibrations, a healing and self-transformative effect can take place. Other artists such as the theorist/composer Dr. Jeffrey D. Thompson of the Center for Neuroacoustic Research have been using computer-modelling and various synthesis/spectral filtering techniques to make audible such phenomena as the magnetic fields of planets and various chaotic processes. Many of the artists within the "stochastic music" movement of the 1950s, of whom Iannis Xenakis was a major exponent, were attempting to use sound to make the complex and esoteric aesthetics of mathematics more accessible on an *experiential* level.

2.2 The Nature of Ritual

Before we continue, it seems pertinent to clarify the definition of the word "ritual" as it will be used in this paper. Unfortunately, the word does carry different meanings for different people, and can suggest rigid, anachronistic religious and social role-plays (for this reason, Pauline Oliveros uses the term "ceremony" rather than "ritual" in her work). Within this discussion, I am using the word "ritual" to describe an activity that incorporates the following features. Firstly, time and space is symbolically defined by the participants and observers. Secondly, an underlying concept is being communicated, usually considered to be important or relevant to the lives of the participants and observers. Thirdly, physical, verbal or sonic metaphors are used to translate the meaning of the concept into a more easily understood form (i.e. to "physically manifest" an esoteric concept that cannot normally be understood). The amount of metaphorical translation can be varied, depending on how important it is for "lay-people" to understand the meaning behind the concepts.

Christopher Small (1998) suggests a slightly different terminology, emphasising the importance of ritual as an affirmation of different levels of *relatedness*:

> *Ritual is a form of organized behaviour in which humans use the language of gesture, or paralanguage, to affirm, to explore and to celebrate their ideas of how the relationships of the cosmos (or of part of it), operate, and thus of how they themselves should relate to it and to one another*. (p. 95)

Small (p. 99) goes on to isolate the term "myth" as being the concept being expressed by a ritual, and "metaphor" (the articulation of the elements and relationships within a myth) being the method of communication of that concept within a ritual. He emphasizes the importance of ritual as an experiential tool for understanding; that the participation in or observation of the ritual process allows the individual

to become part of the symbolic representation of relationships, and thus the experience becomes more intimate than if the concepts were just "explained".

Because of their personal, experiential, and universally shared nature, the physical senses are an effective medium for the metaphorical translation of conceptual meaning. This belief is particularly evident in the complex ritualistic traditions of Tibetan Buddhism, in which meaning is manifested through meditation and ritual using three primary technical aids: "…*yantra, mantra*, and *mudrã*[1]: the parallelism of the visible, the audible and the tangible (i.e., what can be felt). They are the exponents of mind (*citta*) [e.g., mandala symbolism], speech (*vãc, vãcã*), and body (*kãya*)" (Govinda 1960, p. 92).

Yantra is a term that refers to a visual meditational aid for focus and inner clarity. The arrangement of sacred symbols represent a universal set of relationships, and in meditation are either studied or created. *Mantra* is an aural aid. It exists in the form of various sacred seed-syllables that are combined and uttered by the student. These sounds are believed to initiate subtle inner vibrations within the student that resonate with more universal cosmic vibrations. *Mudrã* is a series of bodily (usually hand) movements that are performed during rituals and meditation, and often accompany the chanting of *mantra*. *Mudrã* serves to emphasise the concepts being visualised and vocalised – it reinforces, through association, the power of the symbolism, and makes the meditational process more complete by including the body as well as the voice and the mind in the process.

Govinda asserts that each concept needs to be understood for the action to be effective, and that sound is the vehicle for understanding, not the understanding itself (pp. 93). However, modern artists like Jeffrey Thompson believe that these subtle vibrations are universal and can cause physical and psychological affects that are entirely subliminal. Either way, it is clear that sound can be used as a powerful tool for the realisation of mentally visualized concepts, and that gesture is an excellent method for symbolically "physicalising" these concepts into one's own body, making them more directly experienced. The importance of metaphor and ritual is paramount for these concepts to be effectively communicated.

All of these concepts are directly applicable to interactive music technology, with the added implication that by simply *using* this technology, there can be the implication of certain symbolic associations. When choosing a particular piece of technology, a composer must decide how overtly present (or subtle) the visual and sonic aspects of the "machine presence" should be, and whether this adds or detracts from the audience's musical experience.

Audience members, untrained in the intricacies of computers or composition, may not be interested in the complexities of software design, the sophistication of musical processes, or the amount of memory in a computer. They come to concerts for a variety of reasons, not the least being to enjoy the drama of a live performance. (Winkler 1999, p. 9)

This kind of judgment needs to be equally as sensitive when choosing sounds and instruments. For example, to the Marind-anim people of South-East Asia, a bull-roarer carries a different set of sonic and cultural associations to a drum. Similarly, to a modern Western listener, a synthetic sound will carry a different aesthetic to a raw sample, as much as one of Machover's hyperinstruments would suggest a different kind of visual concept to a totally "novel" controller such as Waisvisz's The Web.

2.3 20th Century Explorations

The 1960s and 1970s was a time of social and political turbulence in much of the Western world. Many European traditions were being challenged, including those artistic and musical. Many artists and composers began to search outside their own cultures to find answers to their questions, with many incorporating philosophies and aesthetics derived from non-Western traditions into their works. Composers such as John Cage, Cornelius Cardew and LaMonte Young began to create works in which the structures, sounds and actions became deliberately charged with symbolic meaning. Indeterminacy and non-linear structures might reflect a Zen-derived concept of time. In Cage's "infamous" 4'33" (1952), ambient sounds become symbols for representing his philosophy regarding the nature of music; in many of the works of LaMonte Young, the role of the audience is altered (sometimes without their

[1] In Microsoft Word, the Anglicised spelling of the Sanskrit words *mudra, vac, vaca* and *kaya* make use of the '~' symbol above the 'a' rather than the correct '-'.

knowledge) in order to challenge their ideas regarding audience/performer relationships. Sometimes, even a musical instrument is used as a symbolic object in a ritual act (e.g. *Piano Piece for David Tudor No. 2* (Nyman 1974, p. 72)).

Although the concepts, methods and motivations of 20th Century avant-garde composers are very different to those of Tibetan monks, the essential tools are very similar. A myth (concept or philosophy) is presented in the form of metaphor (structural, aural and physical representation) to communicate to – and personally involve – the participants (the line between audience/performer start to become blurred). Of course one does not need to be aligned with any particular "non-Western" form of thought to experiment with ritual forms in music. These methods can be used to serve any kind of aesthetic philosophy, and can be either explicit or implicit.

Pauline Oliveros began experimenting with her so-called "ceremonial" works in the late 1970s. She had abandoned standard notation in 1964 in favour of a more theatrical approach (e.g. *Pieces of Eight* (1964), *Aeolian Partitions* (1969)), through which "…she found that her penchant for imagery allowed her to make statements about music that were impossible to say in a totally abstract medium, … and theater pieces provided the opportunity to use material objects to augment her message" (von Gunden 1983, p. 70).

In the early 1970s Oliveros began to study some of the "deeper" aspects of listening and music-making, and became interested in creativity on both metaphysical (dreams, myths, meditation) and physical (Tai-Chi, karate) levels. This co-exploration of the senses culminated in the *Sonic Meditations* (1971), a collection of 12 text works carefully designed to guide the participant into new levels of interplay between two models of information processing – focal attention and global awareness. "Attention is narrow, pointed and selective. Awareness is broad, diffuse and inclusive" (Oliveros 1984, p. 139).

These meditations led to her so-called "ceremonial works" of the later 1970s such as *Crow II* (1976) and Rose *Moon* (1977), in which the fundamental concepts of awareness outlined in the *Sonic Meditations* are used, with the added elements of physical symbolism and ritual action.

The ritual of *Rose Moon* begins with the mandala-style spatial layout, one of the most important visual/conceptual symbols in Oliveros' work. The circular performance space (to be surrounded by the audience) is marked with the points of the compass, and two runners – who alternate every half-hour – circumnavigate the mandala while making a continuous, repetitive percussive sound with bells attached to belts around their waists. 8 musicians sit just within the circumference, and sound their instruments each time the runner passes directly behind their spine. These musicians recite at will the word "moon" in various languages, as well as names of people they feel should be remembered. Audiences may also participate in this activity if they wish.

Inside this circle, 7 other musicians move in procession in the opposite direction to the runner, sounding their ceremonial percussion instruments each time they circle the mandala while following the sound of a "moon rattle", which is passed from player to player and must always be shaken in time with the runners' bells. Like most mandalas, there are segmentations within the circle, and in *Rose Moon* these are divided (using coloured sheets) into 4 triangular quadrants representing each of the 4 elements. These quadrants are occupied by 12 "cloth people" who meditate and sing chords within their 4 groups, while manipulating the sheets in various symbolically choreographed ways. The centre of the mandala is marked with a black and white tent, in which two "moon figures" – one male and one female – enter and remove their clothing.

Rose Moon does not function like a "regular" performance piece, it has no linear narrative. Time is cyclical and multidimensional; there is action and reaction, and the presentation of opposites. Oliveros makes it clear that the work should be performed as an act of highly focussed meditation for both the audience and performers. Much of the significance of this gruelling 2-hour ritual lies in the co-operation of the performers, and in the active involvement of the audience (this is certainly not a work designed for passive entertainment). If the activities are not performed in harmony, the sound is discontinuous and the ritual disintegrates.

The tools of ritual are also strongly represented in *Rose Moon*: the mandala concept is clearly articulated visually (*yantra*) by the placement of the various characters; each set of characters utter different kinds of sounds (*mantra*), giving further definition each group; the sounds and identity of each group or

character are further emphasised by their symbolic actions (*mudrã*).

This model of spatially, aurally and gesturally symbolic performance art could present a vast number of creative possibilities to the modern interactive computer music artist. For example, the role of the audience could be greatly increased by using a video monitoring system such as Eric Singer's Max object Cyclops or STEIM's BigEye. The symbolic sounds generated by the characters within the mandala structure could be changed depending on their (or someone else's) position within the space (e.g. by using movement and proximity sensors attached to an Infusion Systems I-CubeX Digitizer), or by a change of their gesture or its relationship with someone else's. Depending on the software programming, the process could be made as complex or simple as the composer desires, and the sounds much more flexible. This is not to suggest that *Rose Moon* could (or should) be improved, but that the kind of ritualistic structure used in this piece could be a potent and flexible creative foundation for the application of interactive technology.

Karlheinz Stockhausen is another composer who adopted ritualistic multimedia techniques in some of his works during the 1970s. Throughout the 1950s and 1960s, his explorations of musical symbolism were generally confined to "sound in space" as the symbolic medium. Earlier works such as *Gruppen* (1955-57) and *Carré* (1959-60) dealt with the relationships between spatially separated sounds, and those relationships carried much of the symbolic meaning. In the 1960s he began dealing with space in a metaphorical sense, by exploring the relationships between frequency and timbre, pitch and rhythm, or electronic and natural sound. Many of his works during the 1970s extend these ideas further into the realm of fully realised music-ritual works by the addition of visual and physical symbolism. This new development was possibly due to his short period of experimentation with so-called "intuitive" music, in which notation was abandoned in favour of text instructions. The series *Aus den Sieben Tagen* (1968) deals with various aspects of intuitive creativity, meditation and sonic awareness in what Robin Maconie (1976) describes as "…training the mind to prolong the moment of intuition indefinitely" (pp. 173).

In 1970, Stockhausen returned to a more structured composition style in his piano work *Mantra*, but the new importance placed on personal experience, meditation and consciousness in music remained, and he developed these ideas in many ritualistic multimedia works, eventually leading to the monumental *LICHT* opera cycle. Some works, such as *Inori* (1973-4) use gesture as a way of visually symbolizing the sonic and conceptual elements of the music, while others such as *Sternklang* (1971) and *Musik im Bauch* (1975) are fully realised pieces of ritualistic music-theatre.

Musik im Bauch resembles a rite of passage in which 3 "brothers" explore their own melodies, eventually discovering that the same melodies are being played inside the belly of a mannequin, dressed in Stockhausen's clothes, adorned with bells and with the head of an eagle. They open the belly and extract three music boxes, one for each "brother". First they play along with their musical boxes using glockenspiels, but the piece comes to a close with the boxes winding down naturally by themselves, the stage having been abandoned. During the course of the whole piece, 3 other percussionists play stretched out versions of the same musical material, as well as provide cues for sudden actions from the three "actor/percussionists". Through the augmented melodies, two alternate time scales are suggested that create a multidimensional quality to the perceived passage of time within the work. These performers also eventually leave the stage, moving like tin soldiers as the music boxes slow down.

Much of the charm of this work lies in the fact that the meaning is interpretative, giving it an element of "audience interactivity" by allowing each member of the audience to make it "his own" ritual. Unlike Oliveros' *Rose Moon*, *Musik im Bauch* is more dramatic and narrative in that it tells a story, rather than create a meditative cyclical time structure. Maconie's own interpretation is as follows: "The moral seems to be that what we take to be humanly inspired is in fact entirely preordained in the divine melodies of which the composer, in his guise as bird-man, is the mysterious messenger. Music is the ultimate meaning of the visible drama" (p. 246).

Once again, the three-fold method of concept transmission is clear: the mentally visualised concept (*yantra*), demonstrated by the stage layout, multilayered treatment of time and the significance of the visual objects used in the performance (the music

boxes and mannequin); the sonic element (*mantra)*, which is generated by "human" means (in this case by only playing instruments, not singing) or by "automatic" means (music boxes); and the bodily element (*mudrã*), represented by the character of the different acts performed by both the actors and percussionists. Again we can see that the overall meaning is conveyed not by the visual elements, sounds or actions themselves, but by the entire ritualistic process of how the elements interact with each other, and how the audience relates to them.

There are many aesthetic and practical differences between the work of Oliveros and Stockhausen, but they have both made very effective and personally creative use of similar ritualistic elements. In addition, both these composers have and do use interactive technology and live computer applications in their music, and both were composing electronic music as early as the 1950s.

For many years Oliveros has used complex user-controlled delay techniques, both in her own solo performances and in ensemble projects such as The Deep Listening Band (primarily using the "Expanded Instrument System": Lexicon PCM 42 digital delays, Lexicon PCM 70 reverb units, MIDI controlled amplifiers and a Reson8 DSP Processor), and has collaborated with artists such as Doug Van Noort, whose installation using various controllers and Opcode's Max/MSP formed part of her recent work *The Library of Maps: An Opera in Many Parts*.

Stockhausen is also taking advantage of new technology, such as his use of highly complex sound diffusion systems in the more recent developments of the *LICHT* opera cycle.

It is difficult to say, if today's technology was available 25 years ago, whether they would have employed such technology in their ritualistic works of the 1970s, and if so, how different the works would be. Needless to say, the temporal and spatial structures presented by these works, as well as the use of certain "universal" ritual tools and methods, provide us with some fascinating models for application in modern interactive computer music.

3 New Media and Methods

One of the features of modern gesturally and spatially controlled instruments is that they do not follow the same kind of idiomatic "rules" as acoustic instruments: either in the playing technique or non-musical associations (which are usually as a result of the instrument's socio-cultural history).

For traditional instruments, new developments or improvements not only resulted in a complete change to the style of playing and composing, but were also a symbolic representation of the social and cultural climate of the time. It would be impossible to play a Franz Liszt etude on a harpsichord, not only because the instrument would be physically incapable of this feat (due to its narrower range, lighter frame, different sound, etc.), but it would also be *aesthetically* inappropriate. Thus each instrument's identity is defined by its physical capabilities, as well as by its many non-musical implications. "The invention of the modern piano, for example, … fuelled the imagination and technique of nineteenth-century Romantic composers…The aesthetic and the instrument are wedded" (Winkler 1999, p. 313).

Interactive computer music is much less imbued with social and historical associations, simply because the history is much shorter. Although computer music as a general field has carried various associations over the years, it has followed the rapidly changing attitudes towards computers in general. "In a great deal of music since the advent of electronic sound production, the electronic elements have been used to imply the non-human. […] Part of my contention here is that this view of technology is now no longer relevant. Technology is beginning to empower people" (Garnett 2001, p. 21). Today there is much more choice in how the technology will communicate symbolically – its visual and sonic impact can be highly apparent (even intrusive) or so subtle it can be barely noticeable. Generally, these instruments are also more physically and aurally adaptable. Because there is no *direct* correlation between the gesture and sound of the instrument, the composer must "invent" these correlations – choices must be made carefully in order for these correlations to communicate successfully to an audience.

These two general features of identity can vary greatly with different controller instruments, either physically or mechanically. It may or may not have a connection with a traditional instrument (e.g., by its physical appearance, playing style or sound output). Indeed, an instrument may not even have to be associated with a single object, such as the "…new paradigms…" (Tanaka 2000, p. 391) of public multimedia installations and network music.

Because of the "newness" of so many of these instruments, there is still much exploration being done into the essential aesthetic qualities of this kind of music making. In many ways, a full circle has been traversed, beginning with the advent of electronic recording media. "When recording technologies became socially effective, they brought about … a shift from the prominence of music production processes (composition and interpretation) to the prominence of listening activities as cultural experience" (Iazzetta 2000, p. 259). Electronically-generated sound has deepened these tendencies, but as an interesting recent development, new gesturally-controlled technology has once again brought us to music as something physically *experienced*, with the added excitement of new uncharted aesthetic territory to explore.

Atau Tanaka and Tod Machover are two artists who are making many new practical and theoretical innovations in the field of gesturally-controlled music and its application. Although their methods and ideologies are very different, they do represent two sides of the same aesthetic coin.

One of the main differences between these two artists is that Tanaka is a practicing performer specializing generally in one instrument (the BioMuse, by BioControl Systems, Inc.), whereas Machover realizes his art through composition and multiple instrument design.

Atau Tanaka is involved in many projects that are interesting both from the point of view of new instrument aesthetics, as well as within the context of music ritual. I will briefly discuss one of his major preoccupations – his performances using the BioMuse interface, and his performances with the ensemble Sensorband.

Because of his instrumental specialization, Tanaka aims to communicate to the audience through the way he and his instrument communicate with each other. He describes this combination of the "…physical, tactile or sonic and musical…" as a kind of "feedback" between the performer and his instrument. "It is a confidence in his instrument that helps the musician to create a flowing musical dynamic that is conveyed to the audience" (Tanaka 2000, p. 400).

In many ways this idea could be seen in a similar way to the idea of the mind, voice and body as tools in the process of conveying meaning through ritual. However in Tanaka's case, the ritual is experienced by both the performer and the audience. The BioMuse is a highly expressive instrument, sensing muscle tension in the performers arms via two armbands, and thus is very responsive to physical gesture. Non-essential gestures (e.g. head or leg movements) can also be incorporated into the performance to enhance the level of expressivity, but care must be taken that "… the thin line between musical artistry and vain theatre…" (Tanaka 2000, p. 401) is not crossed.

It is in Atau Tanaka's performances with the group Sensorband that the elements of ritual are fully realized. With Edwin van der Heide and Zbigniew Karkowski (who play their own instruments, the ultrasound MIDI-Conductor and an infared sensor "cage", respectively), the ensemble performs highly engaging group-improvisations and compositions in a way that would be impossible on traditional instruments.

There is a mass of computer-generated electronic sounds coming from three musicians on stage. The audience must distinguish who is playing what. At some moments it is clear, and there are other moments where it is unclear. We can play with this ambiguity. It is a kind of natural reaction on the part of the audience to try to make a connection between the physical gesture they see and what they hear. However, to do so is difficult, because these sounds are unknown. (Tanaka, cited in Bongers 1998, p. 18)

Thus a ritual is created by the very "newness" of the media, combined with the communicative relationship between the receptive, actively listening audience and the sensitive, knowledgeable performer.

I think that what we are doing is really without tradition. Or if there is tradition, it would be right at the roots – music as ritual, trance, and pure energy. […] We want to communicate. Our concerts exploit energy, we want the audience to feel like they have just gotten their batteries reloaded. We want them to feel stronger and like better human beings. (Karkowski/Bongers, p. 23)

Tod Machover is also renowned for his work in refining and expanding the subtle expressivity of musical instruments (e.g. his collaborations with performers such as Yo-Yo Ma (1991) in his design of the so-called "hyperinstruments" – gestural controllers designed to work in conjunction with

existing instrument such as the 'cello), and he is also a highly creative designer of completely new instruments and compositional models.

Much of his work involves the creation of instruments designed to be played by non-musicians. As opposed to the hyperinstruments, which are designed to enable professional performers to reach new levels of expressivity, instruments such as the Melody Easel, Gesture Wall and Drum-Boy are designed for ease of use, allowing amateurs to create and shape their own sounds and musical structures without being intimidated by instruments that are difficult to play. "If we could find a way to allow people to spend the same amount of concentration and effort on listening and thinking and evaluating the difference between things and thinking about how to communicate musical ideas to somebody else, how to make music with somebody else, it would be a great advantage" (Machover 1999, cited in Oteri).

Machover has put a lot of thought into how gesture and sound are intuitively joined, and has developed instruments that are designed to be played in such a way. With colleagues Maggie Orth and Gili Weinberg, he has designed musical toys designed for children such as the "Squeezables" and "Stretchables" – instruments made of soft, pliable materials that take full advantage of the malleability of synthesized sound, and thus simulate this flexibility in a fully experiential, tactile way. In 1996 Machover launched the first stage of a large-scale project called the *Brain Opera*, in which hyperinstruments and instruments for amateurs are used in a combination of live performance and real-time Internet activity. In this project, Machover is attempting to find a perfect balance between predetermined structure and performer/participant free will. "My interest lies in understanding the balance between central organization and anarchy – in our minds and in our lives. The Brain Opera is intended to encourage audiences to reflect on this process" (Machover 1996, unattributed interview).

Whereas the realization of Tanaka's artistic vision lies in his sharing a ritualistic experience with his audience, Machover is exploring the possibilities in creating a set of tools and circumstances that provide the audience with the opportunity to create their own musical ritual. "To create precisely the situation where somebody can do something really personal and special and contribute and feel like something wonderful has happened, that's I think a major goal for a certain kind of work that should be done now, and it's very hard to do" (Machover/Oteri 1999).

4 Conclusion

Although there is no one conceptual model for the communication of meaning, there is still the essential quality of *communication*. As human beings, we do not derive meaning from isolated elements, but from the relationships between elements (as applied in the comparative nature of spoken language). It is for this reason that, no matter what technology is being used or what conceptual/aesthetic framework is being communicated, the foundation of that communication lies in our ability to relate to that framework. As physical beings, we experience and relate to the world by sensation, so a metaphorical language built from the senses serves as an excellent format for the contextualization of meaning. This active, sensory format consists of the concept (mind), the sound (voice), and the gesture (body).

Ritual is essentially the communication of a concept through acted-out metaphors, and how those metaphors relate to each other and to the individual. Modern movement-controlled interactive technology is an ideal medium for the realization of these metaphors, in that it is not bound by any particular set of gesture-to-sound relationships. The ritual can be shaped and manipulated by the composer or by the participants to suit the relationships being explored, thus making it a much more personally relevant and participatory experience.

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